A CRM APPLICATION FOR THECHAISHOP&CO

A PROJECT REPORT

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PG DEPARTMENT OF COMPUTER SCIENCE

A CRM APPLICATION FOR THECHAISHOP&CO

User Story

As a sales associate at **TheChaiShop&Co**, I need a CRM platform that helps me manage customer information effectively and improve the efficiency of sales activities.

The system should enable me to easily access, edit, and maintain customer and employer details in a centralized location. It should also allow the classification of customers into **online and offline categories using record types**, helping provide more personalized communication and services based on their purchasing method.

Furthermore, the CRM application should automatically handle **sales-related calculations**, including determining the order quantity and computing the total billing amount. This will help minimize manual errors and ensure smooth and accurate transaction processing.

I also need the system to generate **interactive reports and dashboards** that present sales data in a clear and organized way. These insights will assist in evaluating sales performance and making better strategic decisions.

Additionally, the application should automatically send **birthday greeting emails with special promotional offers** to customers, helping strengthen relationships and encourage repeat business.

1. INTRODUCTION

1.1 Project Overview

The project titled **"Customer Relationship Management System for TheChaiShop&Co"** focuses on developing a CRM solution using the Salesforce platform to improve how the organization manages its customers and sales operations.

This application will help the company efficiently maintain records related to **customers, employers, and enterprise accounts** while automating several sales and communication processes.

The CRM system will also support **data-driven decision-making** by providing meaningful reports and dashboards that summarize sales activities and trends. In addition,

automated notifications will enhance customer engagement by keeping customers informed about special offers and important occasions.

The **user story and main project goals** described in this section serve as the foundation for designing and implementing the CRM application.

Project Objectives

This project aims to develop a CRM solution that will:

- Simplify the management of customer information
- Automatically calculate order quantities and billing totals
- Ensure accurate storage and handling of customer and sales data
- Provide meaningful insights through reports and dashboards
- Strengthen customer relationships through automated email communication

2. IDEATION PHASE

2.1 Customer Problem Statement

PS	Customer	Goal	Challenge	Reason	Resulting Feeling
PS-1	Sales Executive	Manage customer orders and generate bills	Errors frequently occur during calculations	Billing and order quantities are handled manually	Frustrated and inefficient
PS-2	Store Manager	Review sales performance and customer details	Reports are slow and sometimes incomplete	Data is maintained in different documents and systems	Confused and dissatisfied
PS-3	Customer	Get personalized deals and	Messages are not received at the right time	The system lacks automated notifications	Disconnected and unimportant

		important updates			

2.2 Empathy Map Canvas

Key Users

- Sales Staff
- Operations Managers

Problem Statements – TheChaiShop&Co CRM Application

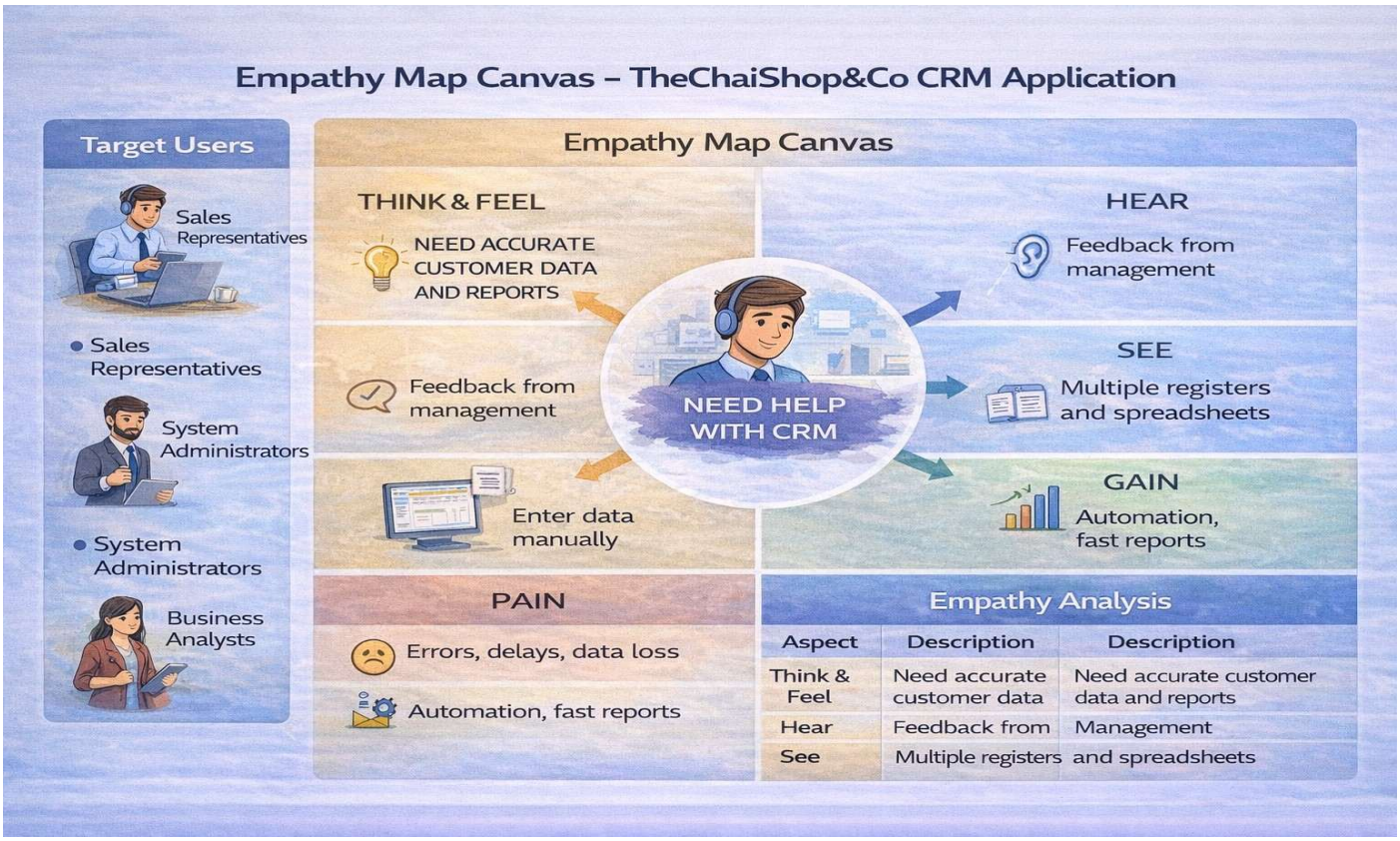
PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1		Manage customer orders and billing	Calculations are often incorrect	Order quantity and billing are manual	
PS-2	Sales Representative	Monitor sales and customer records	Reports are delayed and incomplete	Data is stored in multiple files	Stressed and inefficient
PS-2		Monitor sales and customer ecords	Reports are delayed and incomplete	Data is stored in multiple files	Confused and dissatisfied
PS-3	Customer	Receive personalized offers and updates	I do not get timely notifications	There is no automated communication	Disconnected and undervalued

- CRM Administrators
- Data Analysts

Empathy Insights

Category	Explanation
Think & Feel	They expect dependable customer data and quick access to meaningful reports.
Hear	Managers regularly provide feedback and request performance updates.
See	Data is spread across notebooks, spreadsheets, and manual records.
Say & Do	Users spend time entering and updating data manually in various systems.
Pain Points	Data inconsistencies, slow report preparation, and possible information loss.

Benefits	Automation, faster reporting, and better data organization.



2.3 Brainstorming

Proposed Solutions

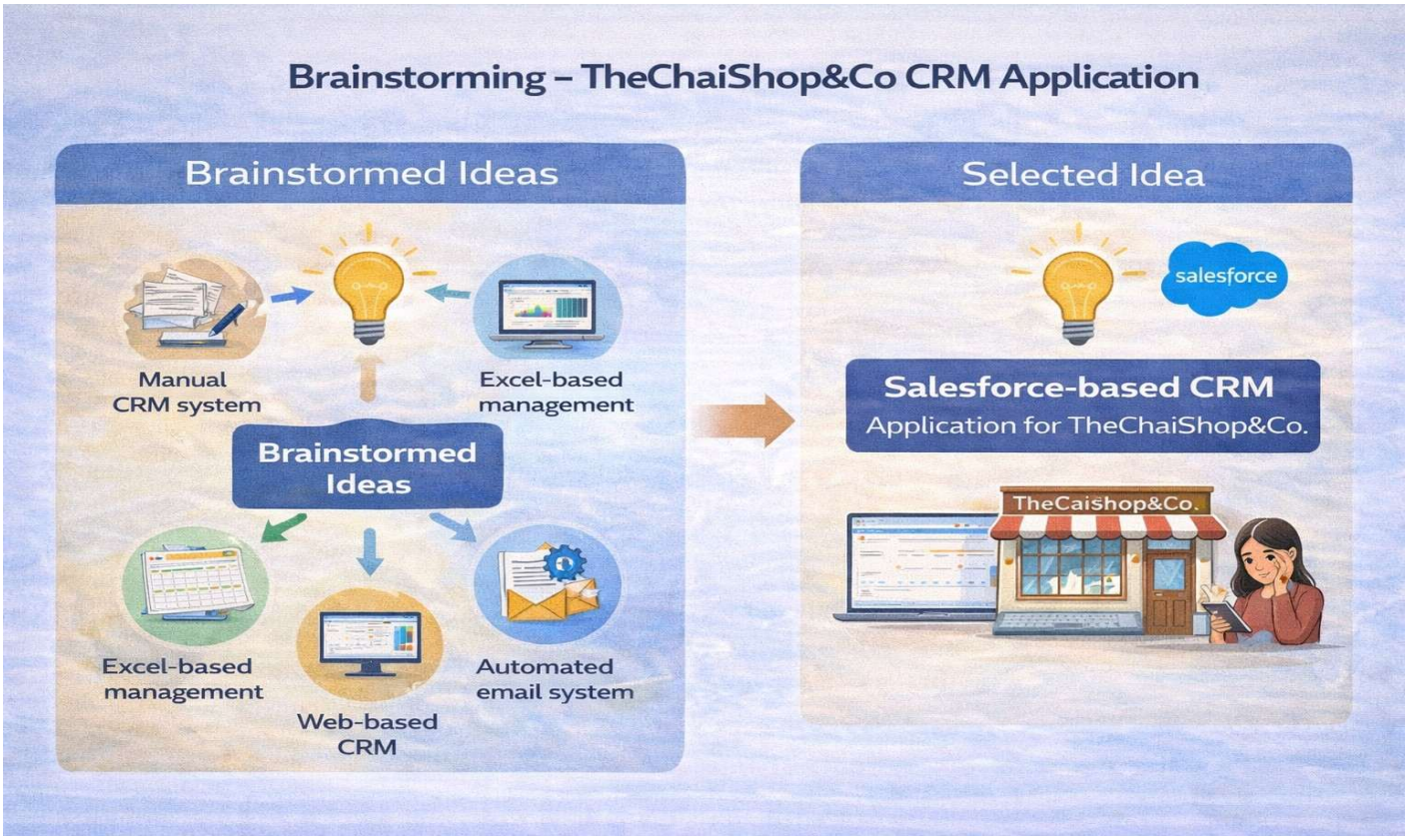
Several approaches were considered to improve customer and sales management:

- Using a **traditional manual CRM process**
- Maintaining customer records through **Excel-based tracking**
- Designing a **custom web-based CRM system**
- Utilizing a **Salesforce CRM platform**
- Introducing **automated email notifications for customers**

Final Selected Solution

Among the proposed options, the most effective solution identified was the development of a **CRM Application using Salesforce for TheChaiShop&Co**.

This approach offers better scalability, automation capabilities, structured data management, and enhanced customer engagement features.



3. REQUIREMENT ANALYSIS PHASE

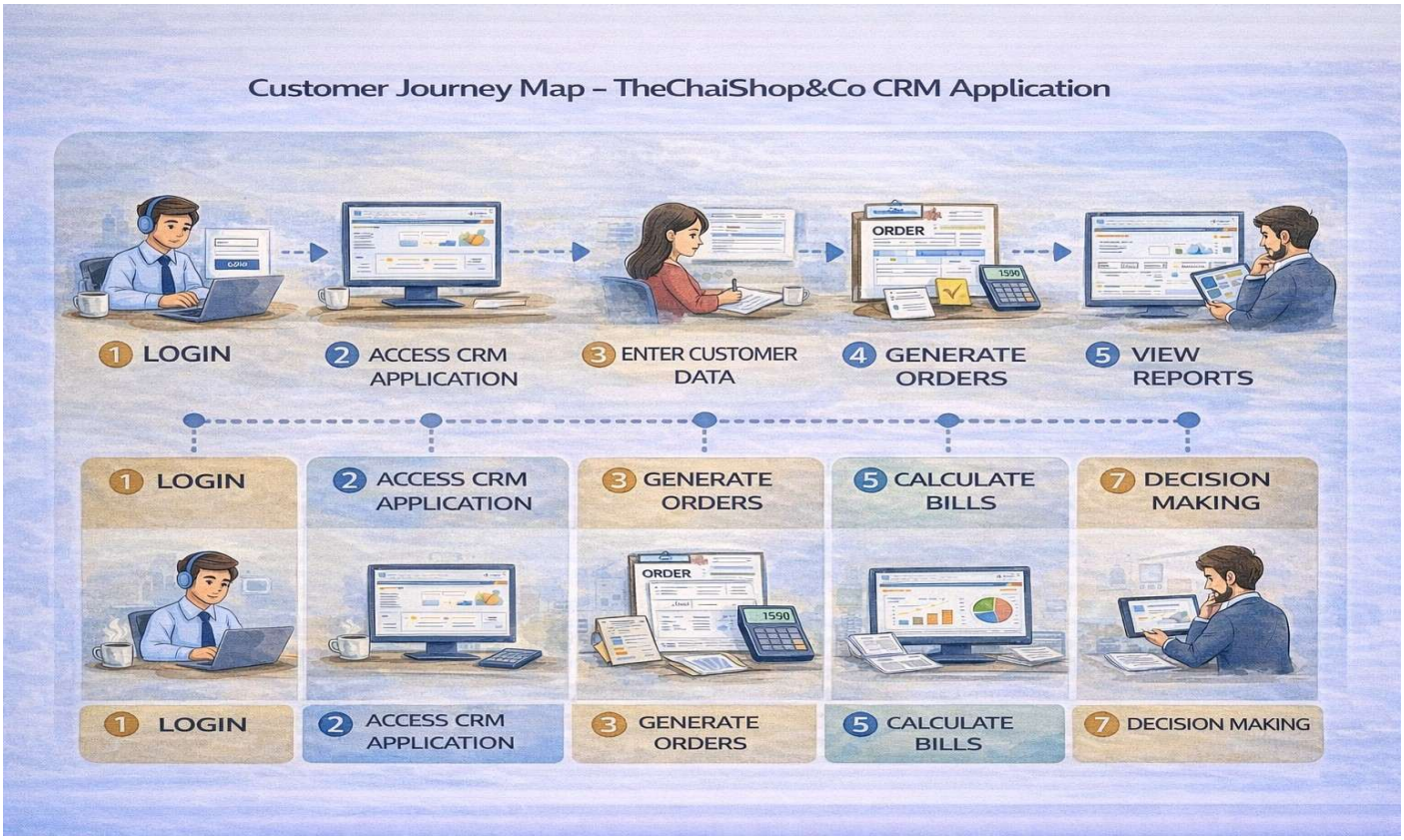
3.1 Customer Journey Map

The CRM workflow illustrates how users interact with the system from accessing the platform to analyzing sales information.

Stages of the Journey

- **System Login** – Authorized users log in to the CRM platform.
- **CRM Application Access** – Users enter the CRM environment to perform tasks.
- **Customer Record Entry** – Customer details are added or updated in the database.
- **Order Creation** – Orders are created for customers within the system.

- **Automatic Bill Generation** – The system computes billing details based on order data.
- **Report Viewing** – Users examine reports and dashboards to understand sales performance.
- **Strategic Decision Making** – Management uses the data insights to plan future business strategies.



3.2 Solution Requirements

Functional Requirements

FR No	Requirement
FR-1	Secure User Login
FR-2	Creation and Management of CRM Objects
FR-3	Entry and Maintenance of Customer Data
FR-4	Automated Order and Billing Calculation
FR-5	Generation of Reports and Dashboards
FR-6	Automatic Email Communication

Non-Functional Requirements

NFR-1: Usability

The system must provide a user-friendly environment so that employees with minimal technical knowledge can easily perform their tasks.

Requirements

- Clear and organized interface design
- Simple navigation between system modules
- Easy data entry forms
- Visual dashboards for quick understanding

Example

A new staff member should quickly learn how to log in, add customer information, and create sales orders without additional training.

Salesforce Lightning interface supports intuitive navigation and customization.

NFR-2: Security

Protecting sensitive customer and company data is an important requirement for the CRM application.

Requirements

- Role-based access restrictions
- Strong authentication for login
- Encryption of stored information
- Backup and recovery of system data

Examples

- Assigning **profiles and permission sets**
- Implementing **object and field level permissions**

- Limiting access to financial and billing records
- Applying secure password and login policies

NFR-3: Reliability

The CRM system should function consistently and ensure that data remains accurate and safe at all times.

Requirements

- Automatic data storage
- Periodic backups of system information
- Proper error detection and handling
- Stable cloud infrastructure

Example

If a user is entering sales data and the connection is lost, the information should be saved and recovered once the system reconnects.

Salesforce cloud services provide reliable backup and recovery features.

NFR-4: Performance

The application should deliver fast system responses even when dealing with large volumes of customer and sales data.

Requirements

- Fast loading of application pages
- Quick generation of reports
- Efficient dashboard performance
- Minimal waiting time for users

Example

When managers generate monthly sales reports containing many customer records, the system should display the results quickly without delays.

System optimization techniques such as caching and efficient queries improve speed.

NFR-5: Availability

The CRM platform must remain accessible to users whenever required.

Requirements

- Round-the-clock system availability
- Minimal maintenance downtime
- Reliable cloud hosting
- Disaster recovery mechanisms

Example

Employees should be able to access the CRM system during busy business hours or outside working hours without interruptions.

Salesforce cloud infrastructure supports high availability.

NFR-6: Scalability

The system should easily support increasing data volume and business growth.

Requirements

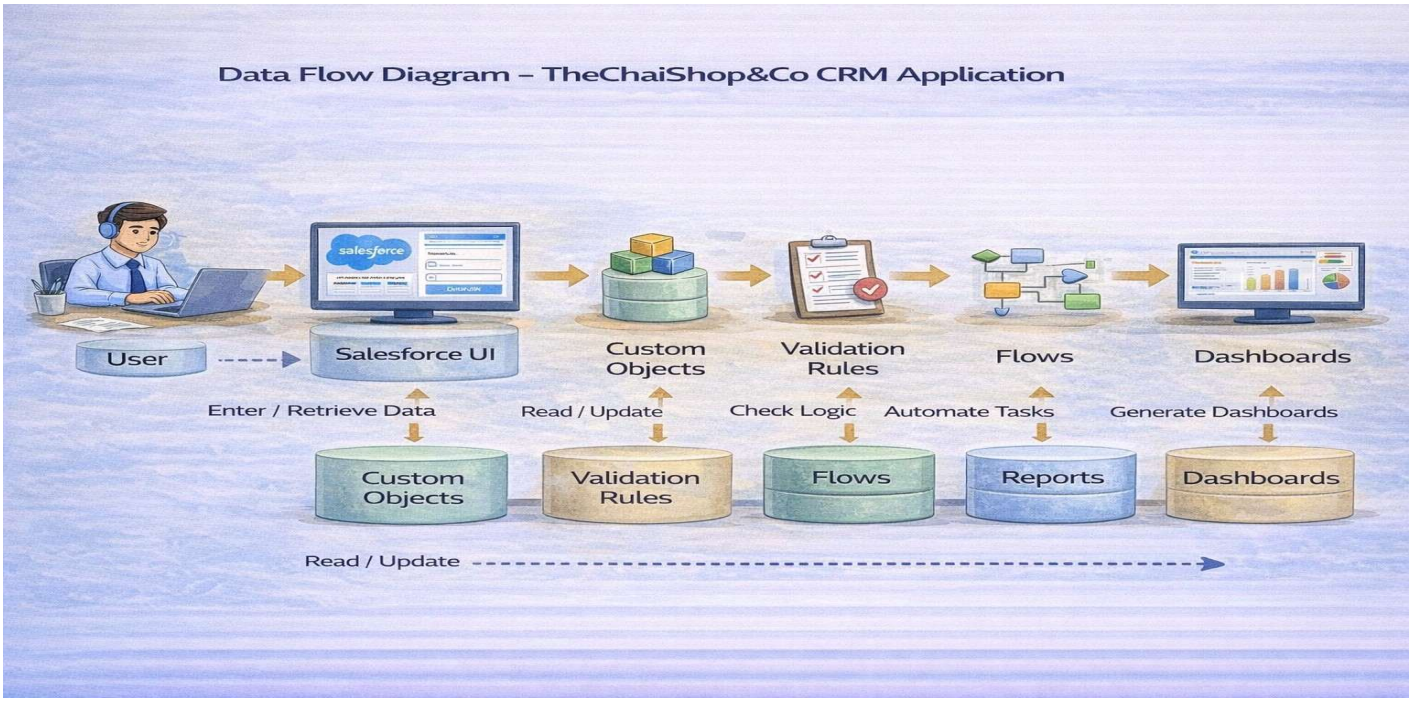
- Ability to accommodate more system users
- Capability to manage larger customer databases
- Expandable storage capacity
- Maintain high performance even with growth

Example

If TheChaiShop&Co expands its operations to multiple locations, the CRM system should still operate efficiently with thousands of customer records.

3.3 Data Flow Diagram

User → Salesforce UI → Custom Objects → Validation Rules → Flows → Reports → Dashboards.



3.4 Technology Stack

Layer	Technology
Frontend	Salesforce Lightning
Database	Salesforce Cloud
Automation	Flows
Backend	Salesforce Platform
Analytics	Reports & Dashboards

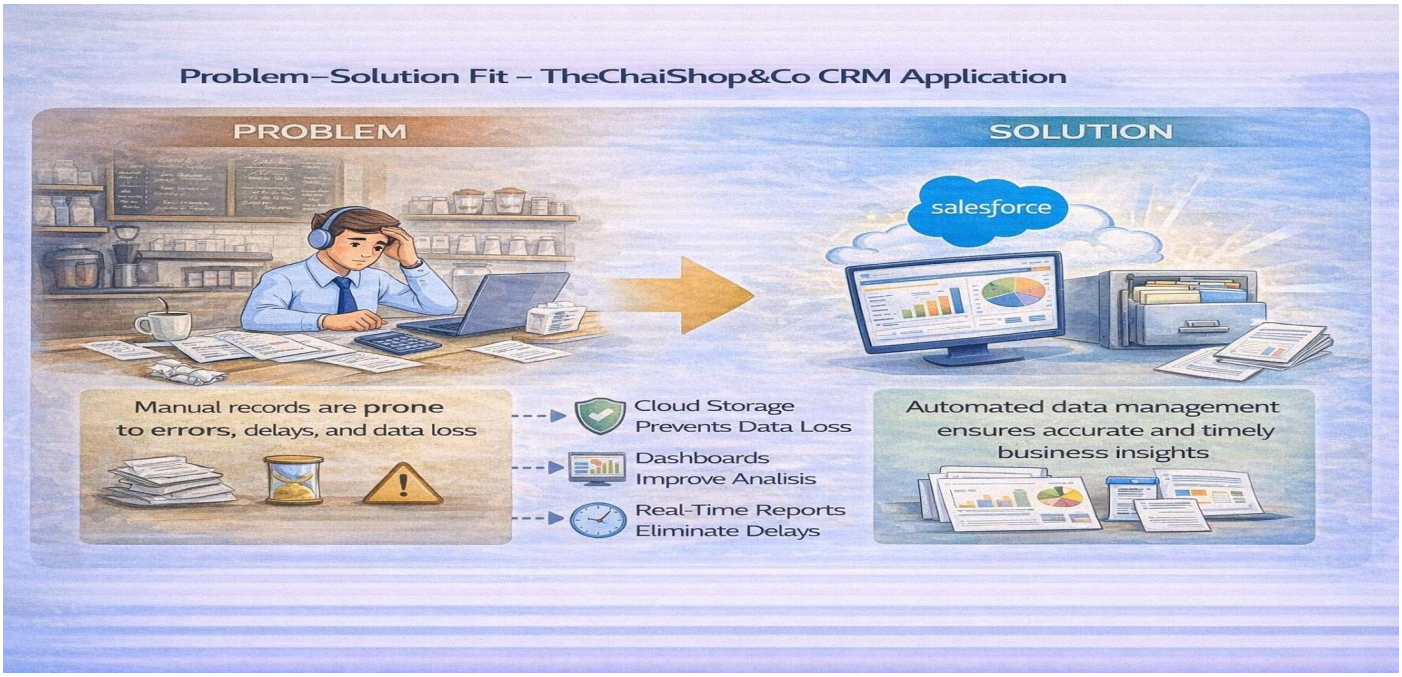
4. PROJECT DESIGN PHASE

4.1 Problem–Solution Fit

The CRM application is designed to solve issues caused by manual data management processes. By shifting to an automated system, customer records, sales details, and billing information can be stored and managed efficiently in a centralized cloud platform.

The use of dashboards and automatically generated reports helps users gain quick insights into sales activities and customer behavior. Additionally, cloud-based storage minimizes the risk of losing important data while allowing easy access from different locations.

This solution ensures improved data accuracy, faster report generation, and better decision-making for business growth.



4.2 Proposed Solution

Functional Requirements

Parameter	Description
User Login and Security	Ensures that only authorized users can access the CRM system through secure login credentials.
Customer Data Management	Allows the system to store and manage customer details including contact information and address.
Employer and Enterprise Accounts	Facilitates the maintenance of employer and enterprise account records for organizational activities.
Order Processing	Supports the creation and monitoring of customer orders with details such as product information and quantity.
Billing Automation	Calculates billing totals automatically using formula fields based on pricing and order quantity.
Data Validation Rules	Prevents incorrect data entry by validating fields such as phone numbers, PAN numbers, and Aadhaar numbers.

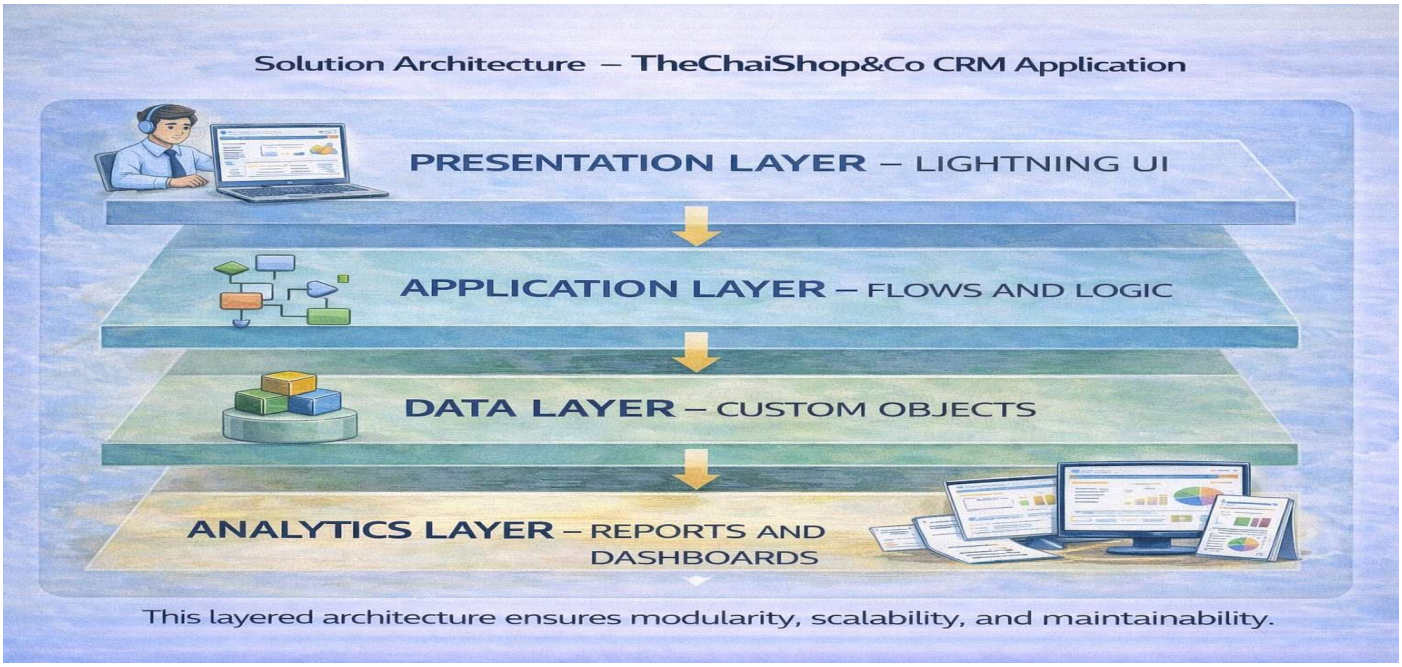
Record Types and Layout Configuration	Provides customized record structures and page layouts for different categories of customers and enterprises.
Global Picklist Values	Ensures consistent field values by using predefined dropdown options for categories, product types, and status fields.
Automated Email Notifications	Sends automatic emails to customers for special occasions, offers, and order-related updates.
Process Automation (Workflow & Flow)	Automates business processes like updating records and generating alerts through Salesforce automation tools.
Sales and Customer Reports	Generates analytical reports that provide insights into sales performance and customer activities.
Interactive Dashboards	Displays graphical summaries of data using charts and dashboards for easy monitoring.
Cloud Data Protection	Safeguards business data through secure cloud storage and provides recovery mechanisms if needed.
User Access Control	Controls user permissions and system access using role-based profiles and permission settings.

4.3 Solution Architecture

Architecture Layers

1. Presentation Layer – Lightning UI
2. Application Layer – Flows and Logic
3. Data Layer – Custom Objects
4. Analytics Layer – Reports and Dashboards

This layered architecture ensures modularity, scalability, and maintainability.



5. PROJECT PLANNING AND SCHEDULING

5.1 Project Planning

- The CRM system was developed using an **Agile development process**, where the project tasks were divided into several sprints.
- A **Salesforce developer environment** was created to begin the configuration and development process.
- **Custom objects** were built to store information about customers, employers, and enterprise accounts.
- **Custom tabs and a Lightning application interface** were implemented to make navigation easier for users.
- The system included **custom fields, relationships, and standardized picklist values** to maintain consistent data.
- **Page layouts and record types** were customized to match different customer categories and user responsibilities.
- **Validation rules** were introduced to ensure users enter correct and complete information.
- **Email automation and flow processes** were configured to send customer notifications automatically.

- **Reports and dashboards** were created to provide insights into sales performance and customer activity.
- Frequent **sprint evaluations and testing phases** ensured progress tracking and issue resolution.
- The project was finished successfully within the scheduled development timeline.

Product Backlog and Sprint Plan

Sprint	Epic	User Story ID	Description	Points	Priority	Assigned To
Sprint-1	Developer Setup	USN-1	As an administrator, I want to set up a Salesforce developer account to start building the CRM system.	1	High	Member1
Sprint-1	Data Structure	USN-2	As an admin, I want to create objects for Customer, Employer, and Enterprise data storage.	3	High	Member2
Sprint-2	Navigation	USN-3	As a user, I want separate tabs for each object to easily access CRM records.	3	High	Member3
Sprint-2	Application Setup	USN-4	As a manager, I want a Lightning App that combines all CRM functions.	3	High	Member4

Sprint-2	Data Fields	USN-5	As an admin, I want to define fields related to customers and orders to record sales data properly.	5	Medium	Member1
Sprint-2	Layout Design	USN-6	As a user, I want customized page layouts for different record types to simplify data entry.	5	High	Member2
Sprint-2	Record Categorization	USN-7	As a manager, I want record types for Retail and Enterprise customers to separate business segments.	5	Medium	Member3
Sprint-3	Validation	USN-8	As a system, I want validation rules to ensure accurate input of customer and order information.	3	High	Member4
Sprint-3	Email Automation	USN-9	As a system, I want flows that automatically send emails to customers regarding offers and updates.	5	High	Member1
Sprint-4	Billing Automation	USN-10	As a system, I want automated calculations for billing amounts using formulas.	5	High	Member2

Sprint-5	Reporting	USN-11	As a manager, I want detailed sales and customer reports for performance analysis.	4	High	Member3
Sprint-6	Dashboard	USN-12	As a manager, I want graphical dashboards that present business data visually.	4	Medium	Member4

Project Tracker & Sprint Progress

Sprint	Story Points	Duration	Start Date	Planned Completion	Points Completed	Actual Completion
Sprint-1	20	6 Days	24 Oct 2022	29 Oct 2022	20	29 Oct 2022
Sprint-2	20	6 Days	31 Oct 2022	05 Nov 2022	20	05 Nov 2022
Sprint-3	20	6 Days	07 Nov 2022	12 Nov 2022	20	12 Nov 2022
Sprint-4	20	6 Days	14 Nov 2022	19 Nov 2022	20	19 Nov 2022

6. PROJECT DEVELOPMENT PHASE

Project Milestones

The project development followed a structured flow through multiple milestones:

- **Milestone 1: Developer Account** – Initialize Salesforce developer environment.
- **Milestone 2: Object Creation** – Build objects for customers, employers, and enterprises.
- **Milestone 3: Tabs** – Add separate tabs for each object for organized access.

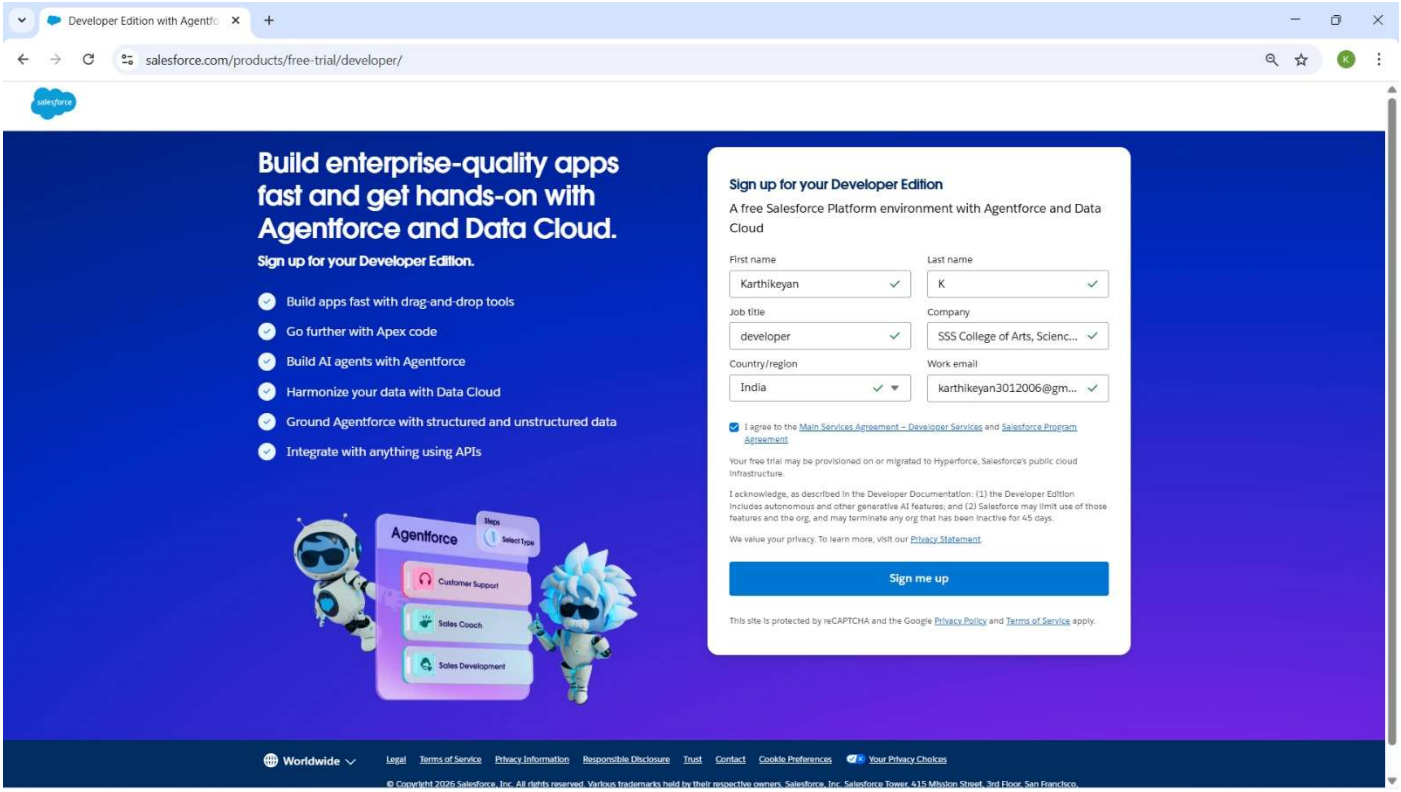
- **Milestone 4: Lightning App** – Develop a single interface for all CRM functionalities.
- **Milestone 5: Global Picklists** – Create standard value sets for fields like product type and status.
- **Milestone 6: Fields & Relationships** – Configure data fields and link objects logically.
- **Milestone 7: Page Layouts** – Customize layouts to display relevant fields for each user.
- **Milestone 8: Record Types** – Implement different record types for Retail and Enterprise customers.
- **Milestone 9: Validation Rules** – Add checks to prevent incorrect data entry.
- **Milestone 10: Email Templates** – Design templates for customer notifications and promotions.
- **Milestone 11: Flows** – Automate routine tasks and notifications.
- **Milestone 12: Reports & Dashboards** – Generate analytical reports and visualize data.
- **Milestone 13: Project Completion** – Final testing, review, and deployment.

Learning Outcomes

Participants will gain practical knowledge and skills in:

1. Executing a **live Salesforce project** with hands-on experience.
2. Creating **standard and custom objects** to store business information.
3. Using **picklist value sets** for consistent data entry.
4. Understanding **field types and relationships** between objects.
5. Designing **page layouts** for better user interaction.
6. Managing **record types** for different customer or business categories.
7. Applying **validation rules** to ensure data integrity.
8. Building **email templates** for automated communication workflows.
9. Implementing **flows** to automate CRM processes.
10. Developing **reports and dashboards** to provide actionable business insights.

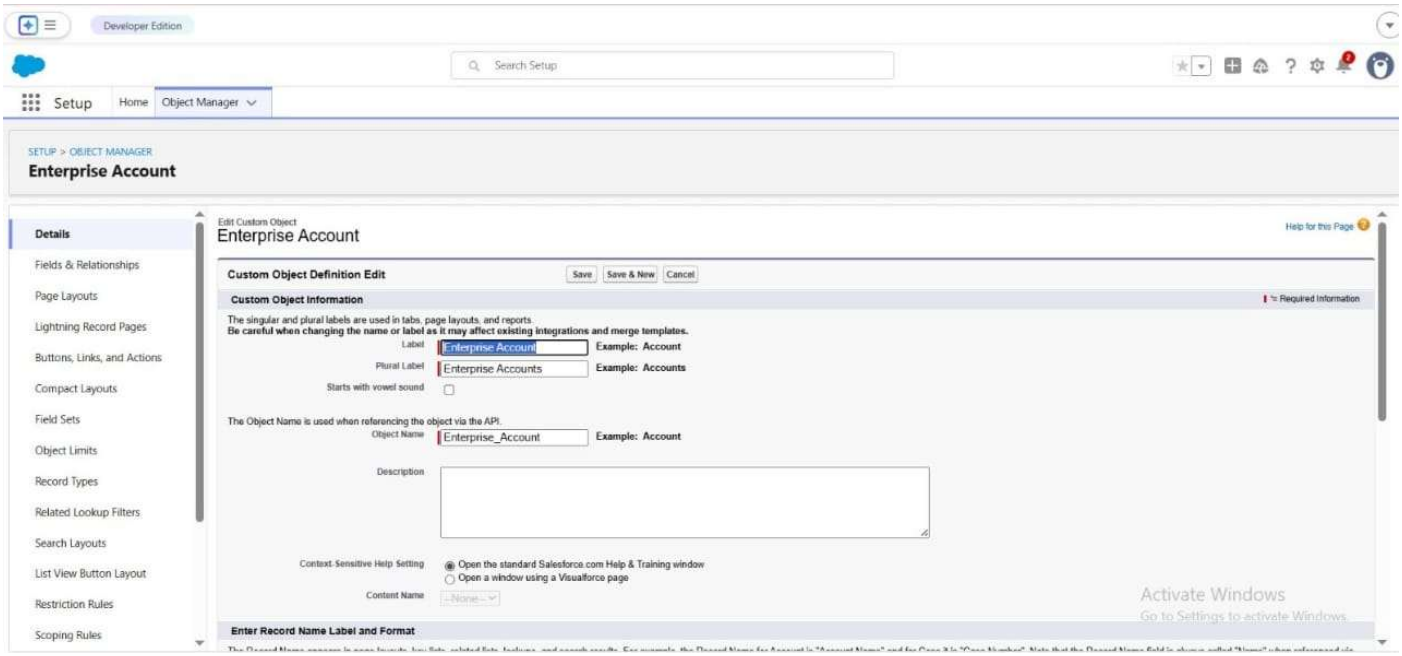
Milestone 1: Developer Account



To begin the project, a **Salesforce developer account** was created. This provided access to the tools and for building the CRM system.

Milestone 2: Object Creation

In this milestone, **objects were designed** to hold customer, employer, and enterprise information. This step ensured that all business data could be stored systematically within Salesforce.


Developer Edition
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Object Manager
SETUP > OBJECT MANAGER
Enterprise Account
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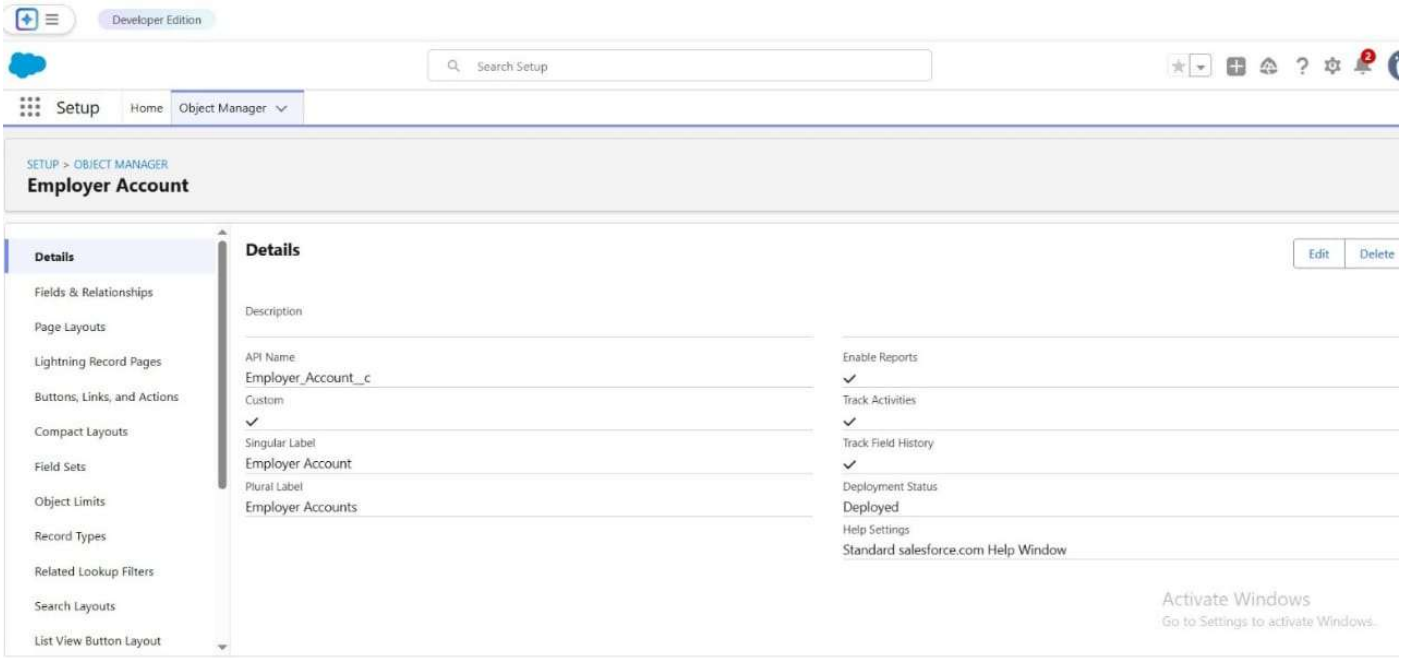
Fields & Relationships
Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets
Object Limits
Record Types
Related Lookup Filters
Search Layouts
List View Button Layout
Restriction Rules
Scoping Rules
Edit Custom Object
Enterprise Account
Help for this Page
Custom Object Definition Edit
Save
Save & New
Cancel
Custom Object Information
= Required Information
The singular and plural labels are used in tabs, page layouts, and reports.
Be careful when changing the name or label as it may affect existing integrations and merge templates.
Label
Enterprise Account
Example: Account
Plural Label
Enterprise Accounts
Example: Accounts
Starts with vowel sound
The Object Name is used when referencing the object via the API.
Object Name
Enterprise_Account
Example: Account
Description
Context-Sensitive Help Setting
Open the standard Salesforce.com Help & Training window
Open a window using a Visualforce page
Content Name
--None--
Enter Record Name Label and Format
Activate Windows
Go to Settings to activate Windows.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Field Sets
Object Limits
Record Types
Related Lookup Filters
Search Layouts
List View Button Layout
Details
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Delete
Description
API Name
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Custom
Singular Label
Employer Account
Plural Label
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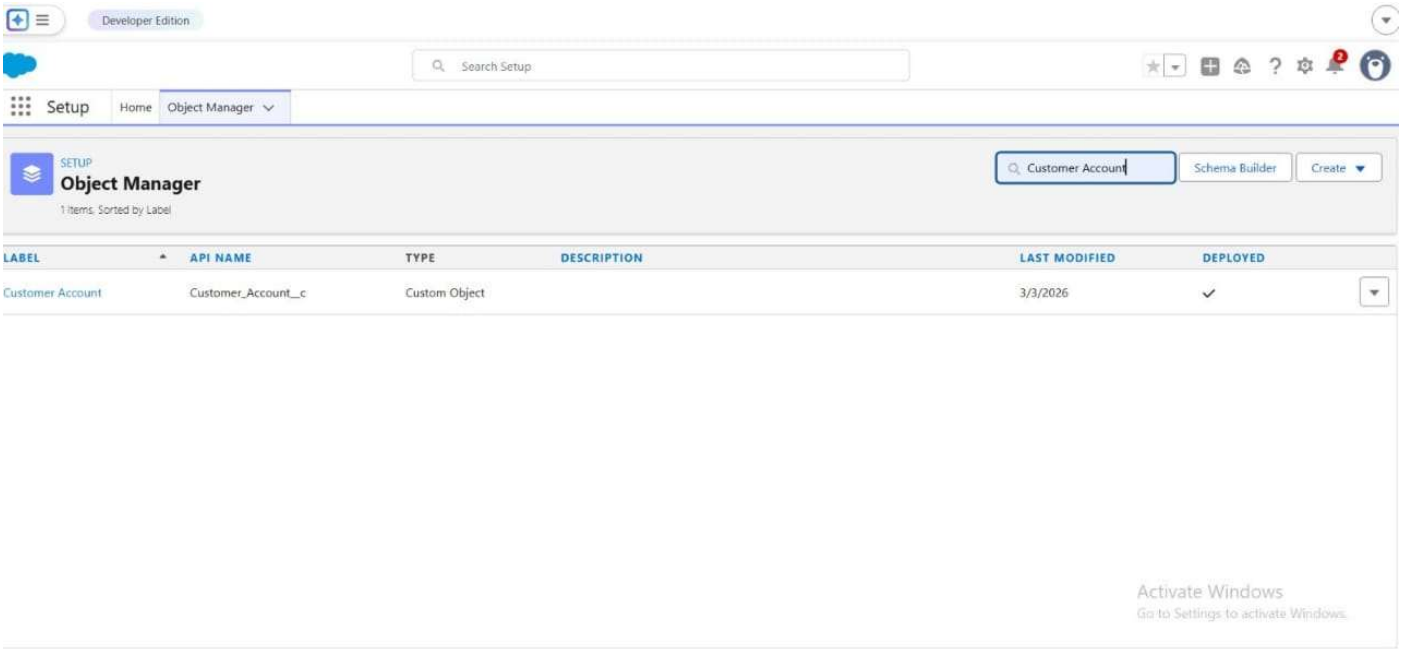
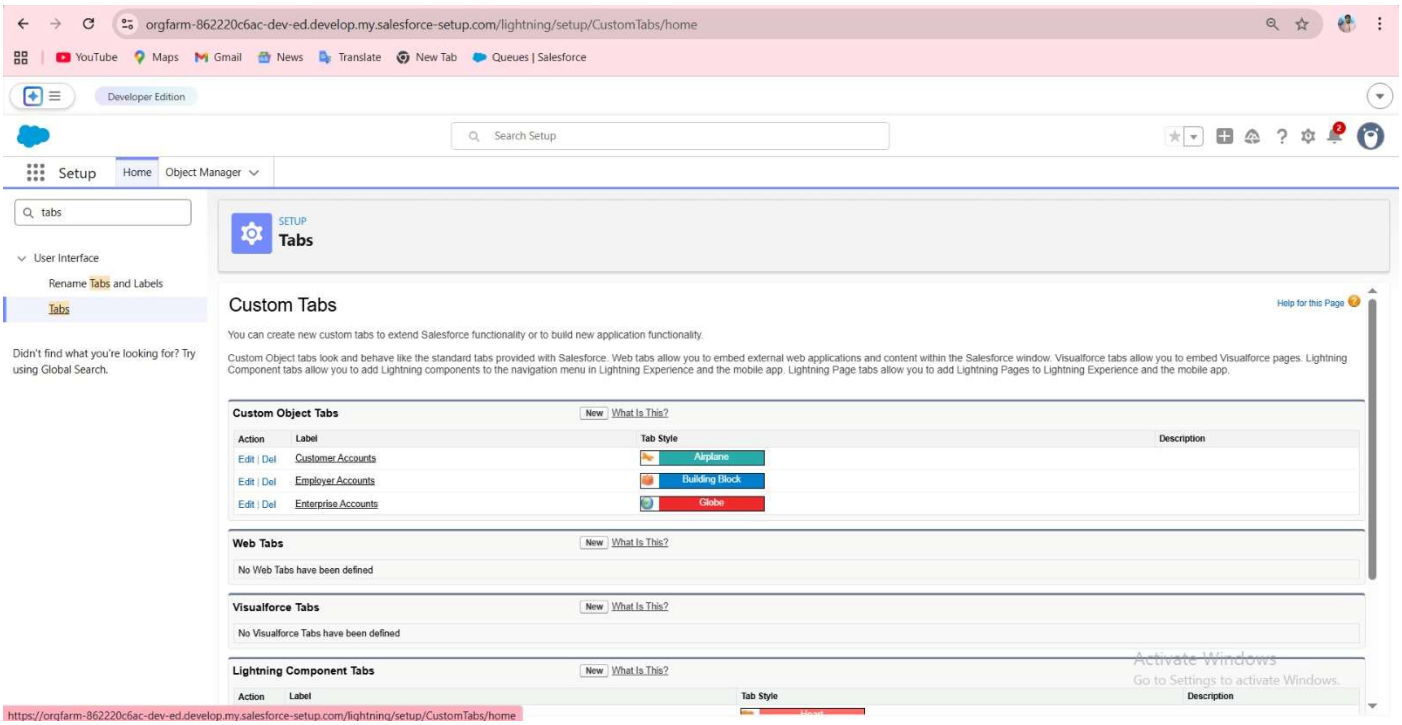
Go to Settings to activate Windows.



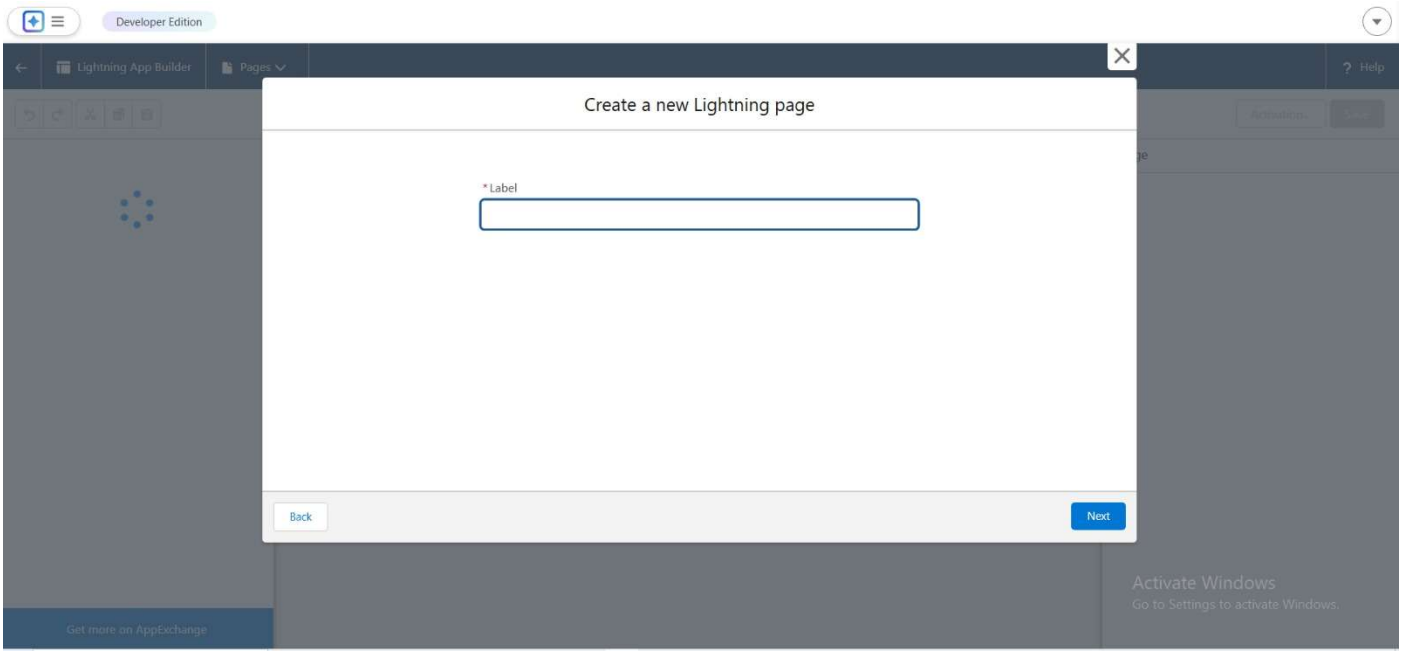
Milestone 3 – Tabs

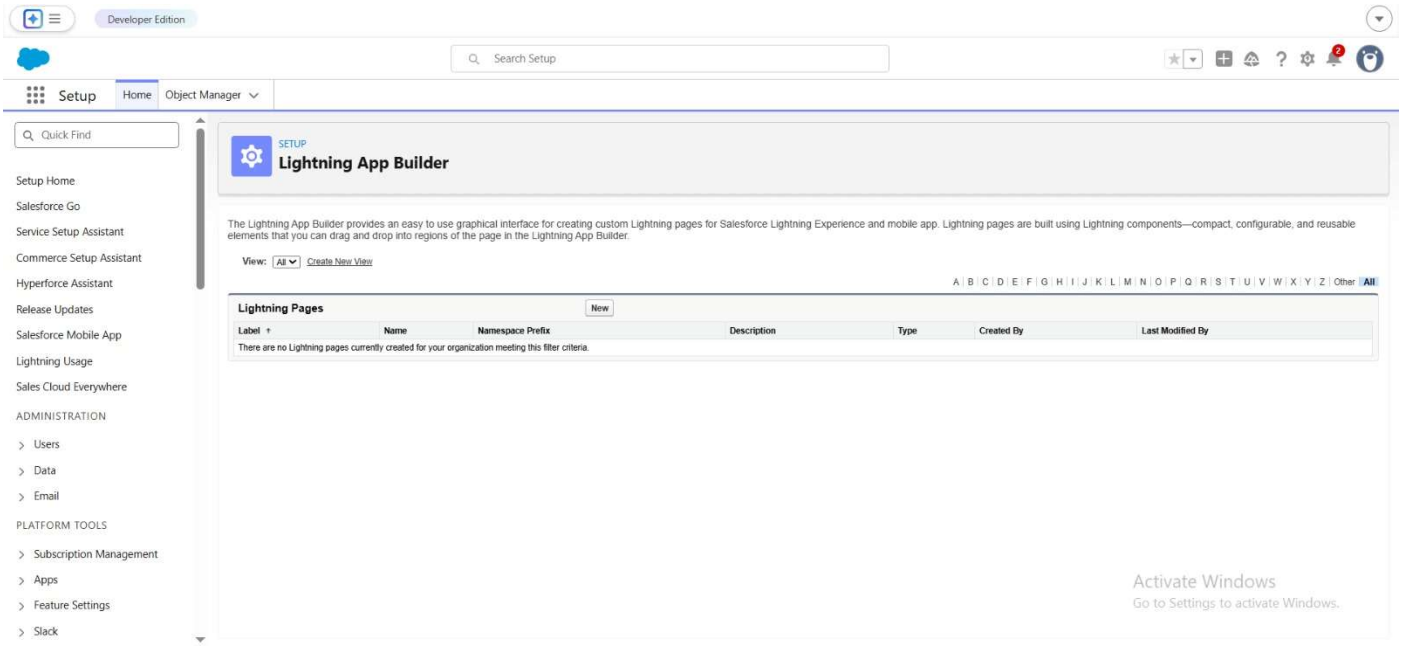
In this milestone, **individual tabs** were set up for all objects. This provided organized access and simplified navigation for users working with different CRM data categories.



Milestone 4 – The Lightning App

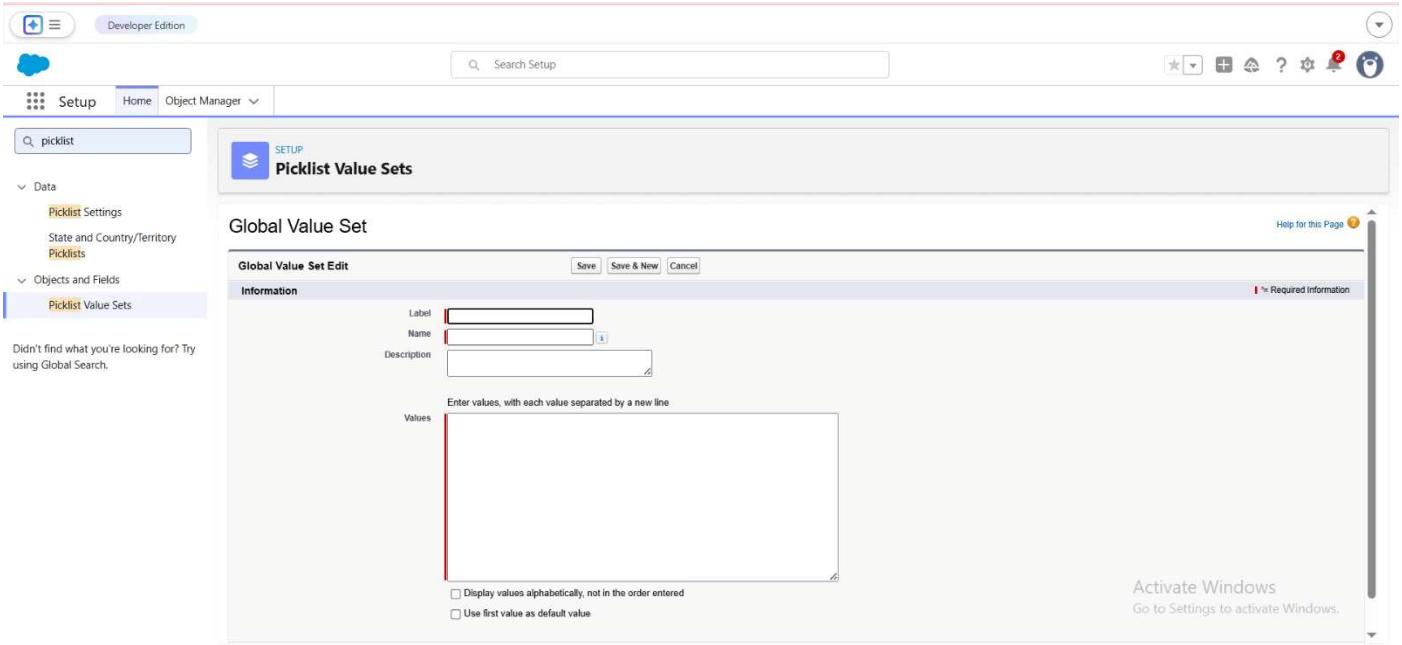
The fourth milestone focused on developing a **Lightning App** to centralize all CRM operations. This provided users with a single dashboard for accessing customer, employer, and enterprise features efficiently.





Milestone 5 – Global Picklist

In this milestone, **standardized picklist values** were created for commonly used fields. This helped maintain uniform data entry across customer, employer, and enterprise records.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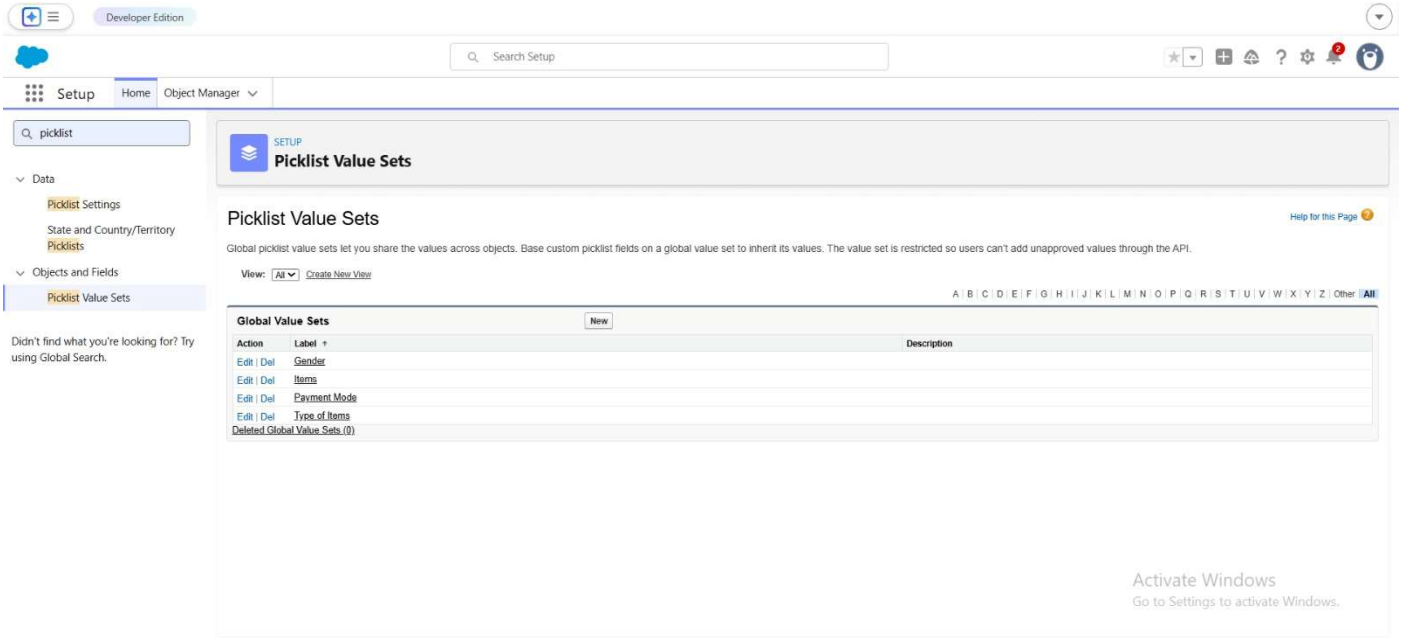
Picklist Settings
State and Country/Territory Picklists
Objects and Fields
Picklist Value Sets
Didn't find what you're looking for? Try using Global Search.
SETUP
Picklist Value Sets
Global Value Set
Help for this Page
Global Value Set Edit
Save
Save & New
Cancel
Information
Required Information
Label
Name
Description
Enter values, with each value separated by a new line
Values
Display values alphabetically, not in the order entered
Use first value as default value
Activate Windows
Go to Settings to activate Windows.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Data
Picklist Settings
State and Country/Territory Picklists
Objects and Fields
Picklist Value Sets
Didn't find what you're looking for? Try using Global Search.
SETUP
Picklist Value Sets
Picklist Value Sets
Help for this Page
Global picklist value sets let you share the values across objects. Base custom picklist fields on a global value set to inherit its values. The value set is restricted so users can't add unapproved values through the API.
View: All
Create New View
A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | Other | All
Global Value Sets
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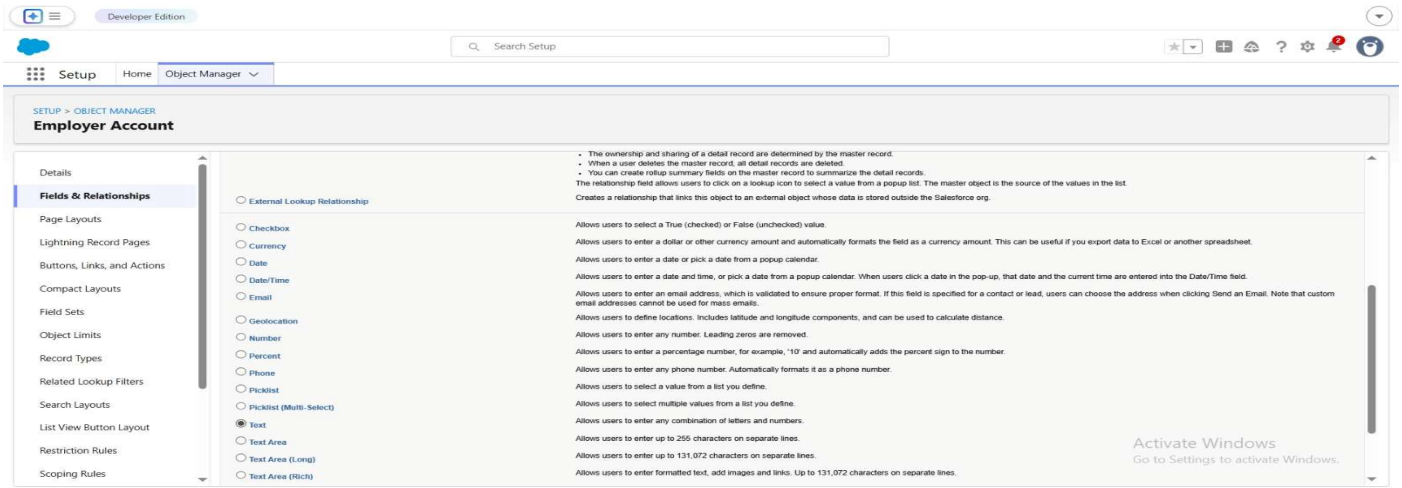
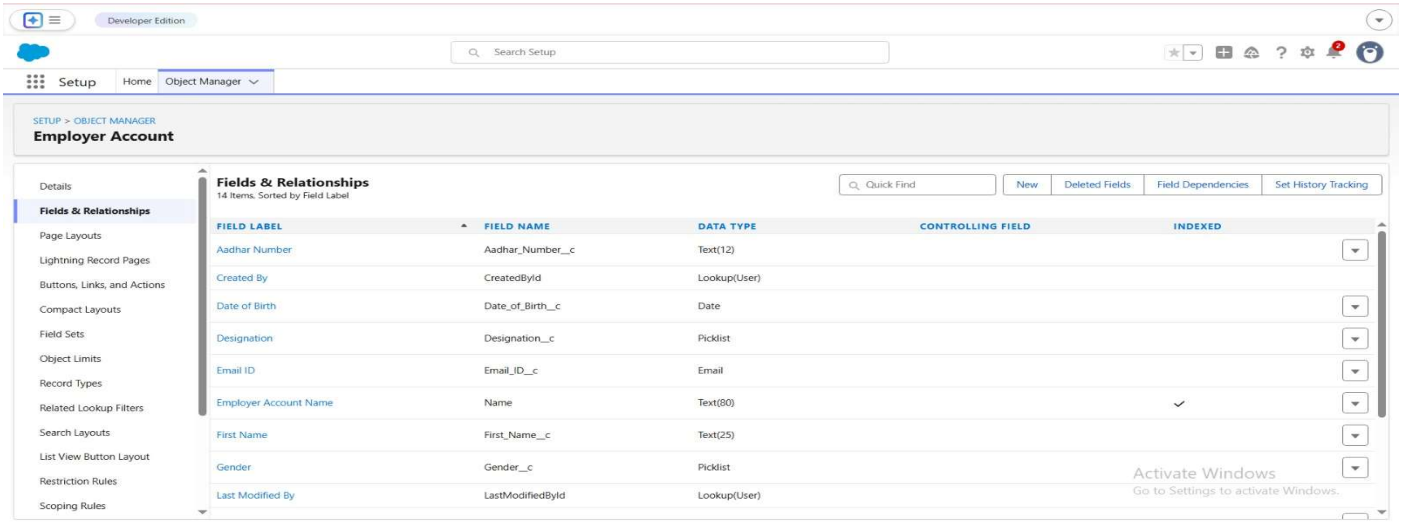
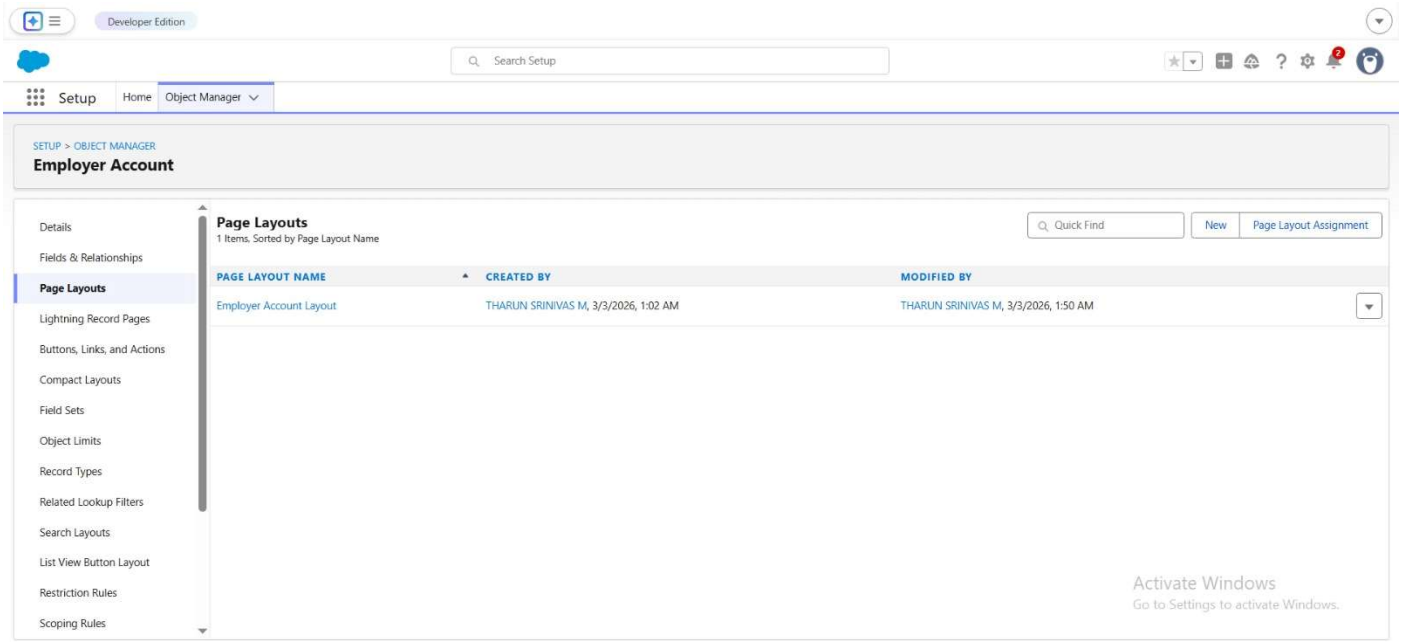
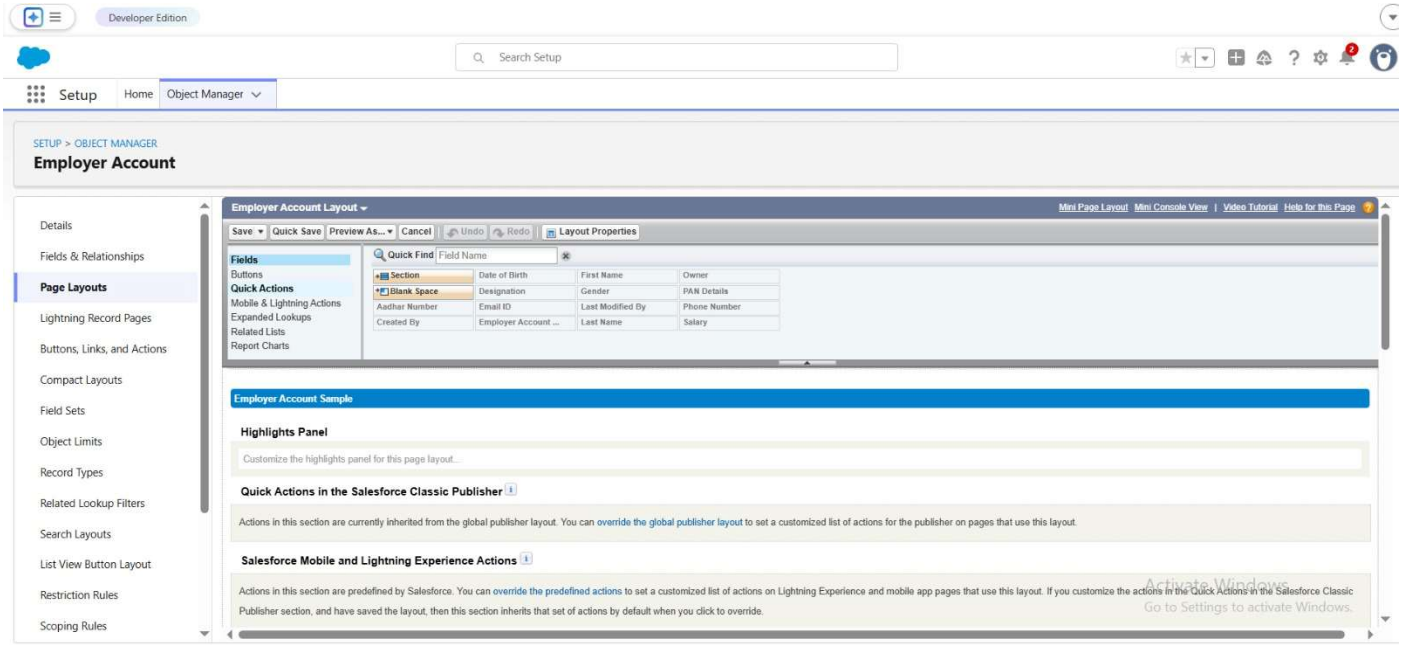
Milestone 6 – Fields and Relationships



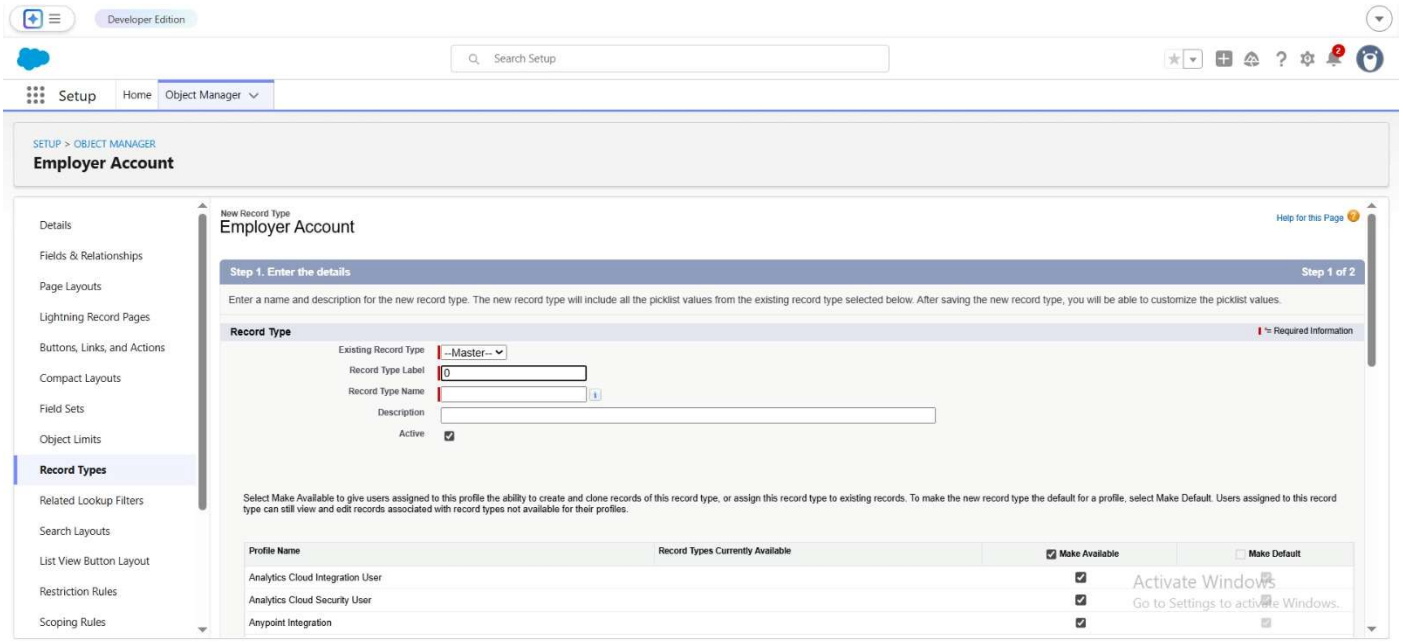


Milestone 7 – Creation of Page Layouts

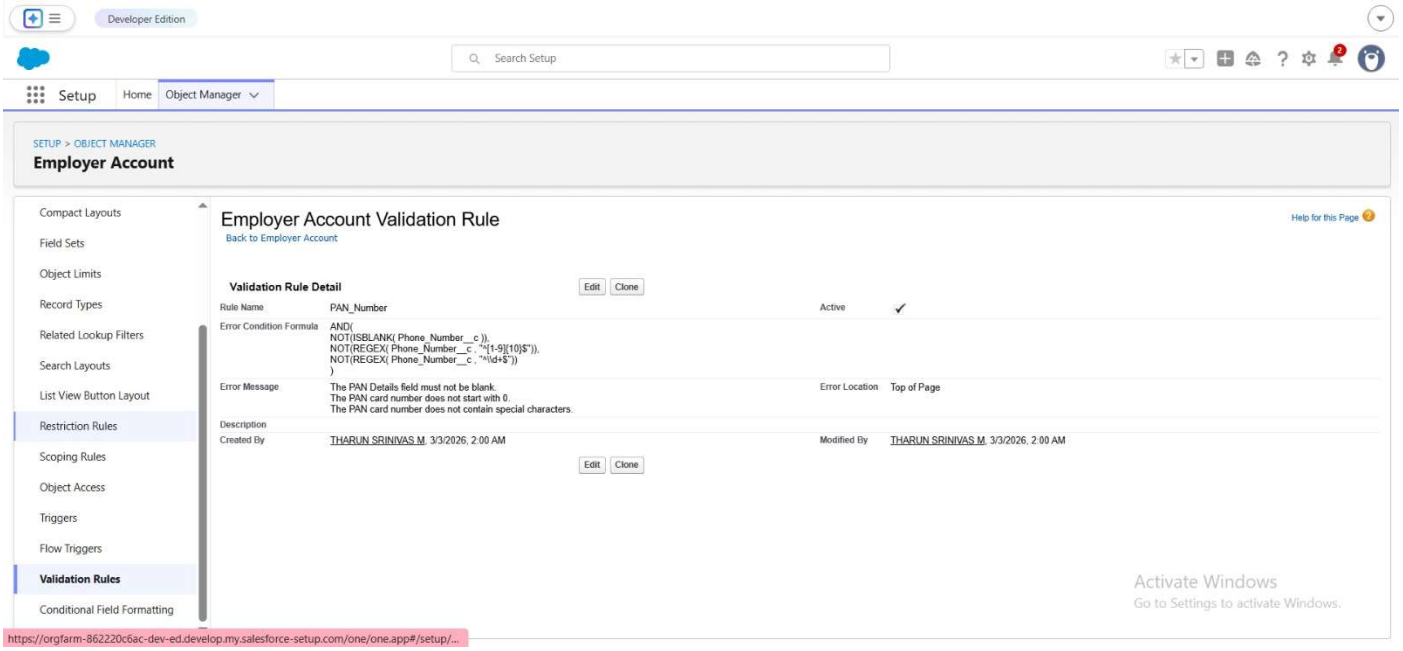


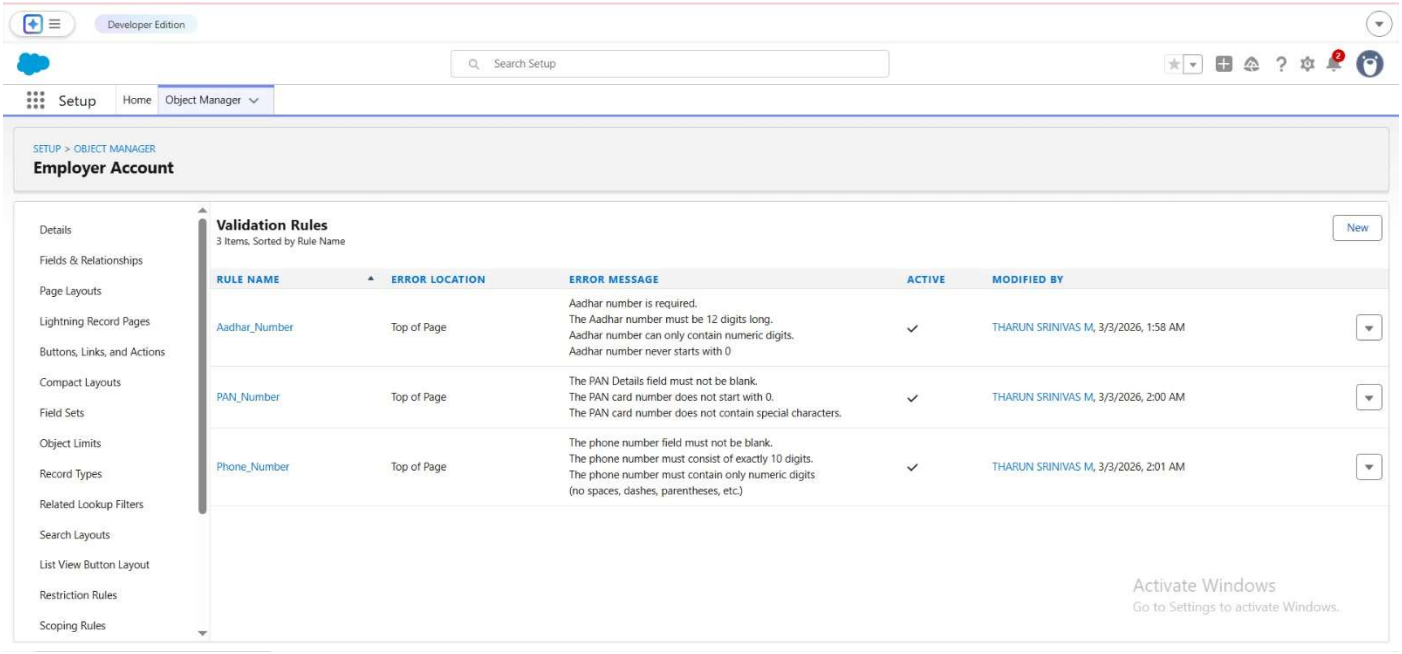


Milestone 8 – Creation of Record Types



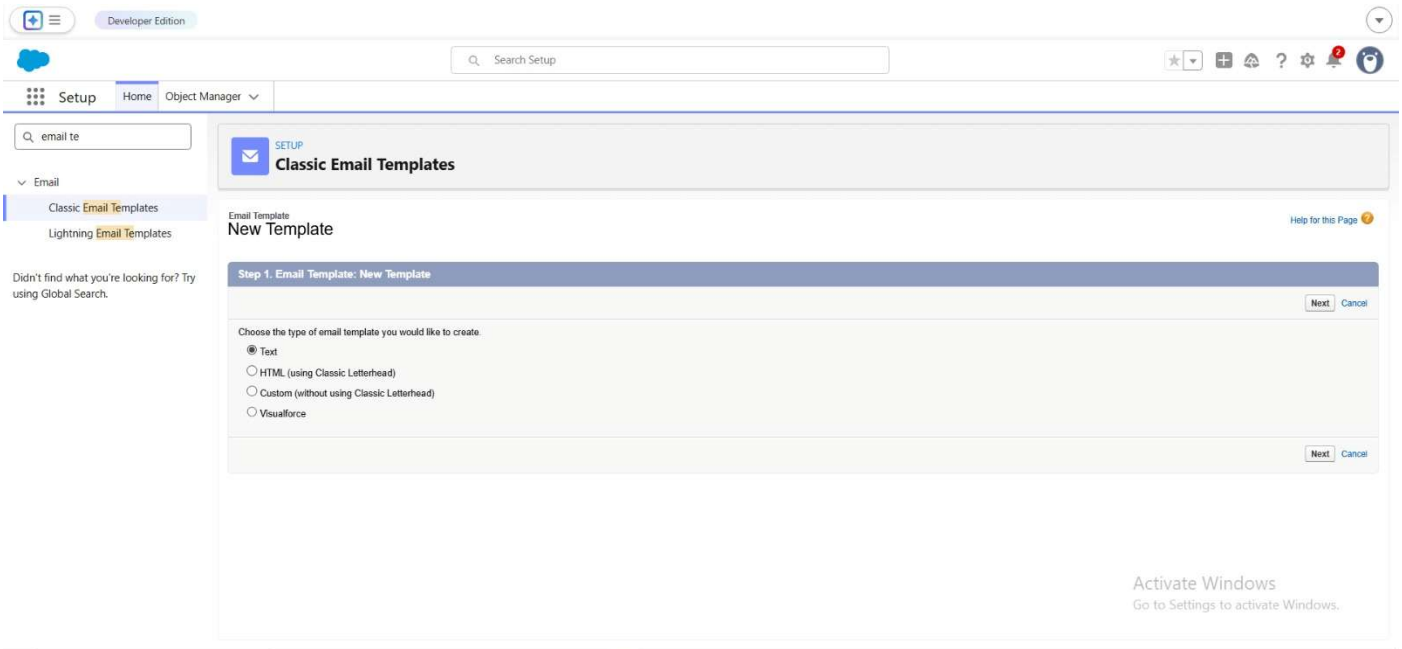
Milestone 9 – Validation Rules

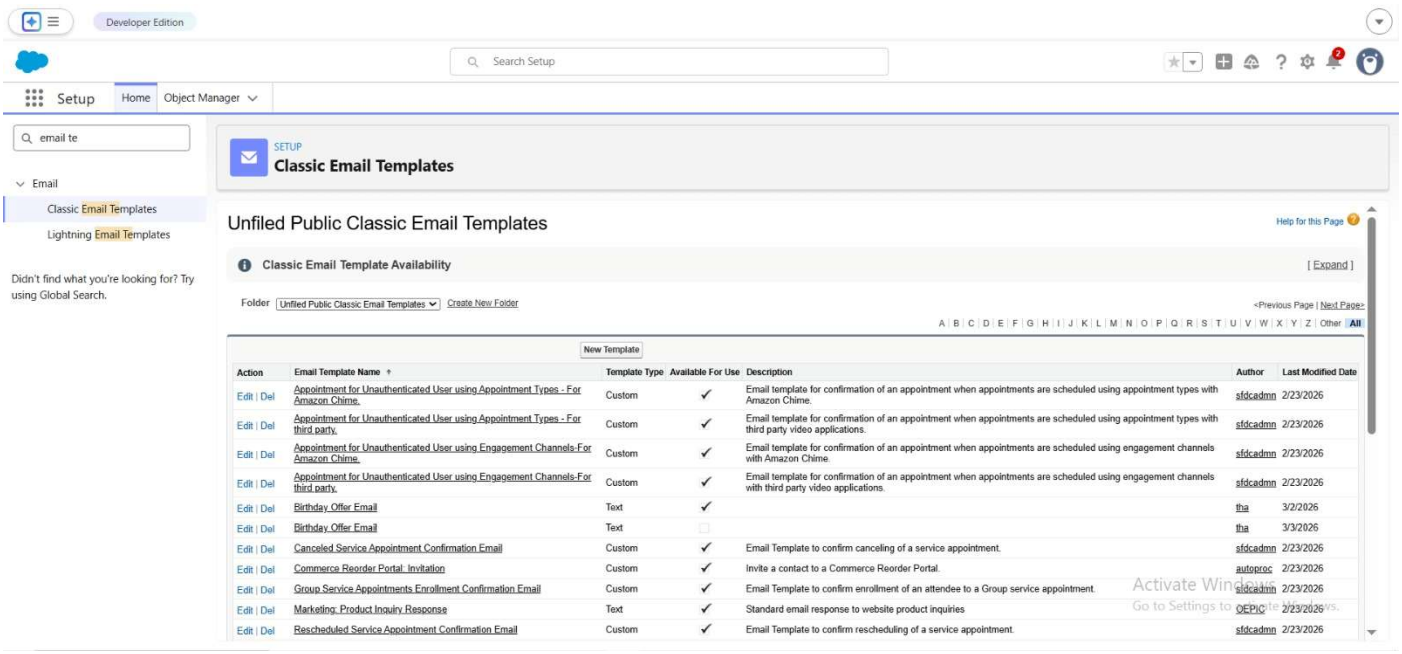




Milestone 10 – Email Templates

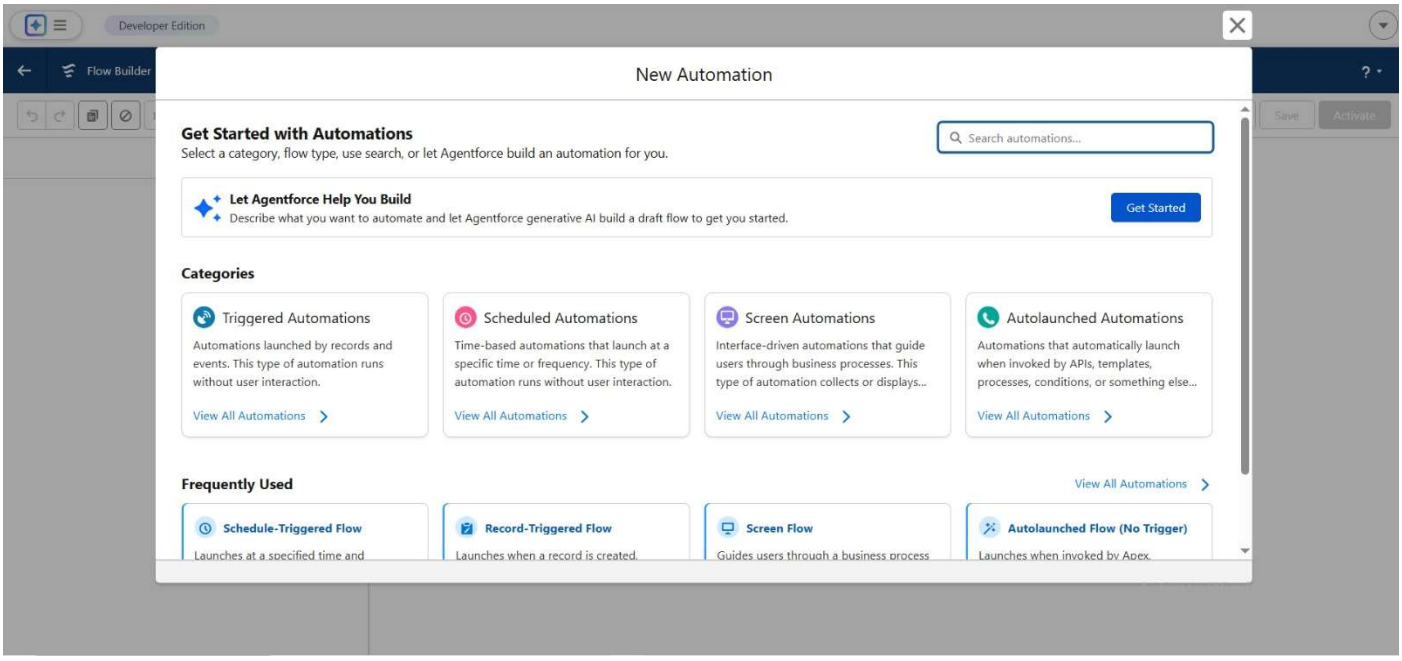
The tenth milestone focused on creating **email templates** for automated customer communication. These templates were used for sending birthday greetings, promotional offers, and order confirmations.





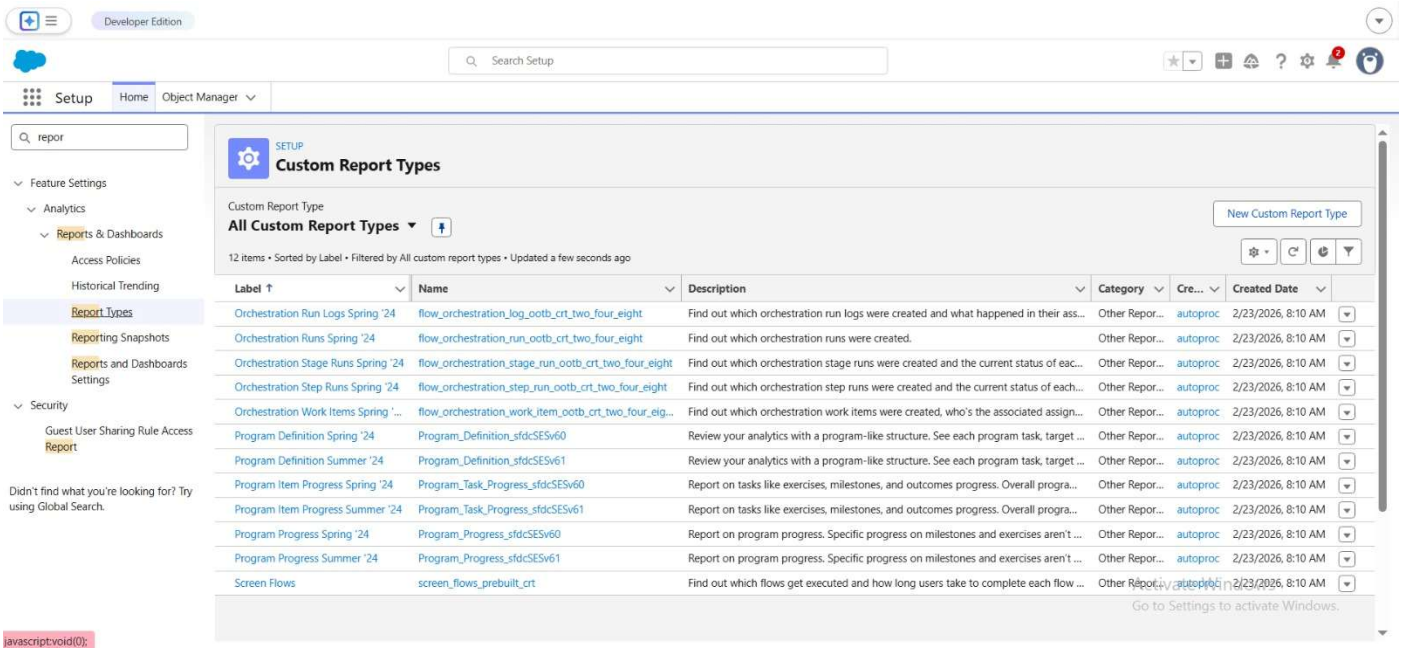
Milestone 11 – Flows

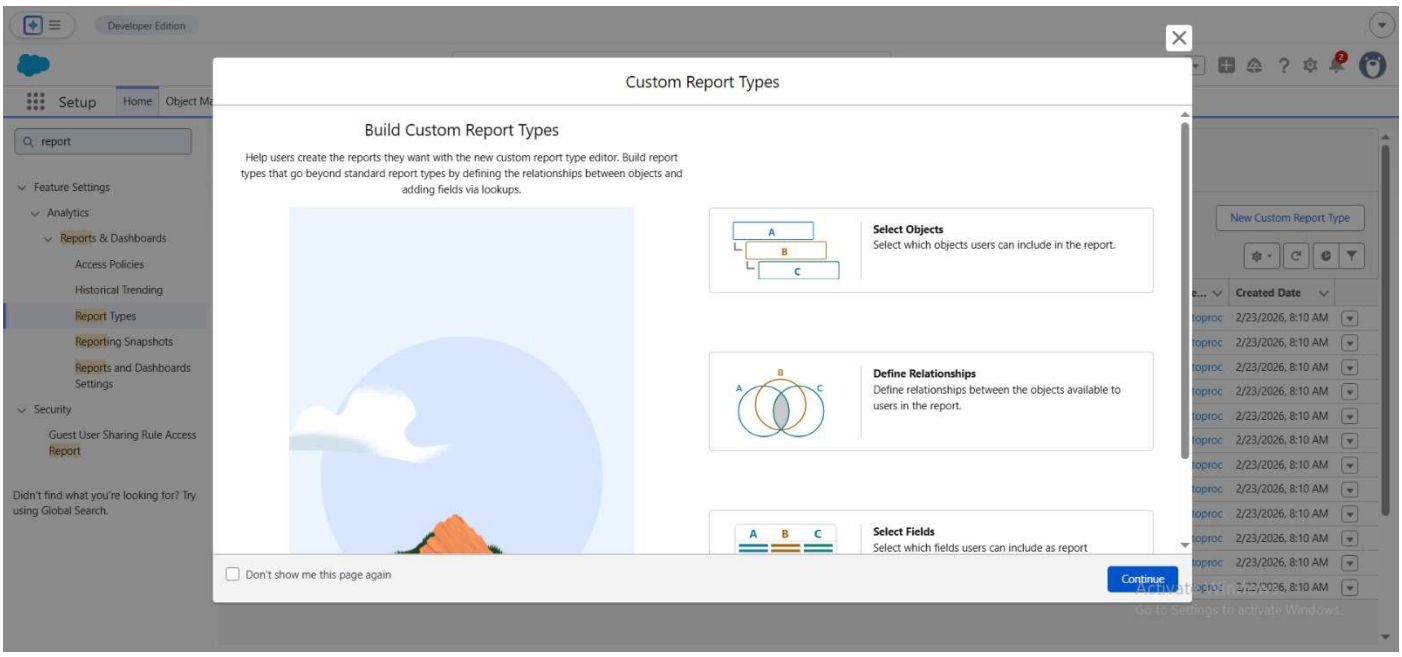
In this milestone, **automated flows** were implemented to manage CRM processes. They ensured that routine tasks like notifications and record updates were performed without manual intervention.

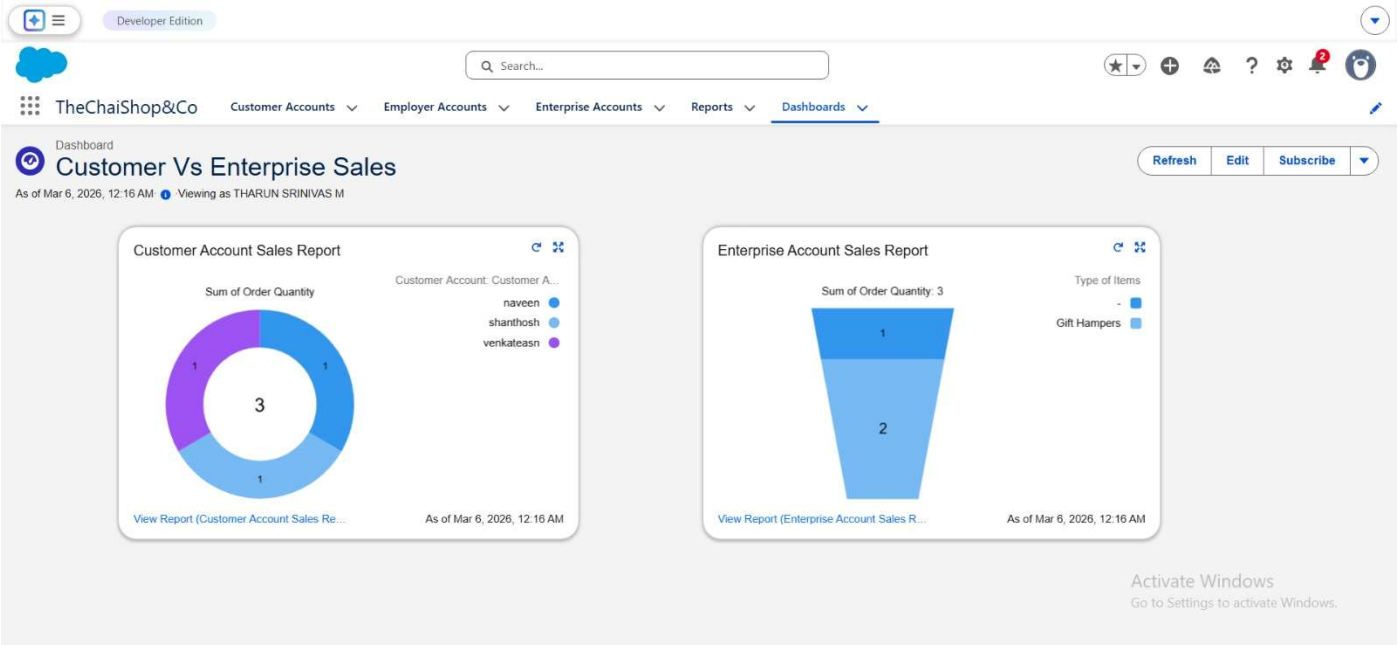


Milestone 12 – Reports and Dashboards

The twelfth milestone focused on creating **reports and dashboards** for real-time insights. These tools allowed managers to monitor sales, customer activity, and enterprise performance effectively.







Milestone 12 - Conclusion

In conclusion, the development of TheChaiShop&Co CRM application marks a significant stride in enhancing the store's operational efficiency and customer relationship management capabilities. Seamlessly integrating employer details, customer information (both online and offline), and enterprise data, this application streamlines business processes and enables informed decision-making. The automation of sales calculations, including order quantity and total bill amount, reduces errors and empowers employees to focus on delivering exceptional service. Moreover, automated email notifications for customers' birthdays foster personalized communication and enhance customer engagement.

7. FUNCTIONAL & PERFORMANCE TESTING

During the testing phase, screenshots were captured from the Salesforce Lightning interface to verify correct system behavior. These include:

- Record creation for Employer, Customer, and Enterprise objects
- Validation rule testing for Aadhar, PAN, and phone numbers

- Verification of record types and page layouts
- Formula field validation for order quantity and billing
- Flow execution for birthday email alerts
- Report and dashboard generation

Performance testing ensured fast response time and system stability during concurrent usage.

8. ADVANTAGES & DISADVANTAGES

Advantages

- Centralized data management
- Reduced manual work
- Automated calculations
- Improved customer engagement
- Real-time analytics

Disadvantages

- Requires Salesforce license
- Internet dependency
- Limited offline access

9. Conclusion:

The CRM Application for TheChaiShop&Co successfully improves business operations by integrating customer data, automating billing, and enabling intelligent reporting. The system reduces errors, saves time, and enhances customer relationships through automation and analytics.

10. FUTURE SCOPE

- Mobile CRM integration
- AI-based sales prediction
- WhatsApp/SMS alerts
- Inventory management
- Multi-branch support
- Advanced customer profiling

11. APPENDIX

Source Code : Salesforce Configuration & Flows

Dataset : Customer and Sales Records

Tools Used : Salesforce Developer Org

Thank You